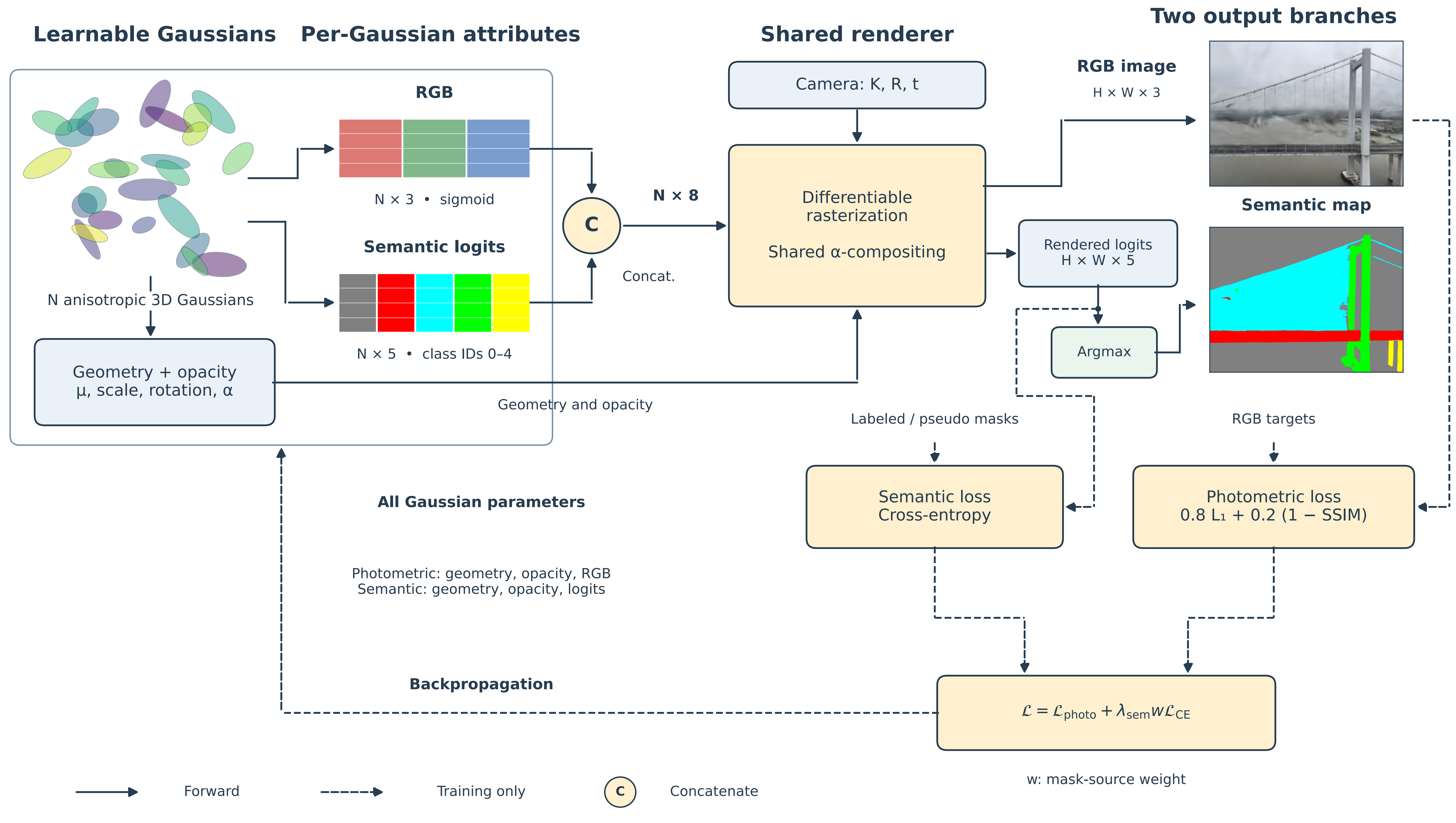

Learnable Gaussians
Per-Gaussian attributes
Shared renderer
Two output branches
RGB
N × 3 • sigmoid
Semantic logits
N × 5 • class IDs 0–4
N anisotropic 3D Gaussians
Geometry + opacity
μ, scale, rotation, α
C
Concat.
N × 8
Camera: K, R, t
Differentiable rasterization
Shared α-compositing
Geometry and opacity
RGB image
H × W × 3
Semantic map
Rendered logits
H × W × 5
Argmax
Labeled / pseudo masks
RGB targets
Semantic loss
Cross-entropy
Photometric loss
0.8 L1 + 0.2 (1 − SSIM)
All Gaussian parameters
Photometric: geometry, opacity, RGB
Semantic: geometry, opacity, logits
Backpropagation
$\mathcal{L} = \mathcal{L}_{\text{photo}} + \lambda_{\text{sem}} w \mathcal{L}_{\text{CE}}$
w: mask-source weight
Forward
Training only
C
Concatenate